Student Number

PEM

2026 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Software Engineering

Total Marks – 80

General Instructions

- Reading time - 5 minutes
- Working time - 2 hours 20 minutes
- Write using black or blue pen
  Black pen is preferred
- Draw diagrams using pencil

Section I Pages 3-6

20 marks

- Attempt Questions 1–16
- Allow about 20 minutes for this section

Section II Pages 7-11

60 marks

- Attempt Questions 17 – 33
- Allow about 2 hours for this section

Directions to School or College

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Section I

20 marks
Attempt Questions 1-16
Allow about 20 minutes for this section

Choose the best answer for the Questions 1-14

1. What is the primary purpose of SQL? (1 mark)

   A. Creating objects in programming languages
   B. Designing user interfaces
   C. Mapping classes to tables
   D. Querying and managing relational databases

2. An organization is upgrading its payroll system and wants to minimize risk during the transition. They decide to run the new system alongside the old system for a period, comparing outputs before fully switching over.

   Which system implementation method are they using? (1 mark)

   A. Direct Installation
   B. Parallel Installation
   C. Phased Installation
   D. Pilot Installation

3. A telecom company wants to predict whether a customer will leave the service based on features like monthly charges, contract type, and customer tenure.

   Which machine learning algorithm is most suitable for this task?
   (1 mark)

   A. Linear Regression
   B. Logistic Regression
   C. Decision Tree
   D. K-Means Clustering

4. What is a key contribution of back-end web development to a web application? (1 mark)

A. Creating page layouts
B. Designing logos and graphics
C. Managing data and server-side functionality
D. Styling buttons and fonts

5. Which of the following best describes the primary purpose of the Domain Name System (DNS)? (1 mark)

A. It translates IP addresses into MAC addresses
B. It converts domain names into IP addresses
C. It encrypts data for secure transmission
D. It manages email delivery between servers

6. Why is stakeholder feedback essential when evaluating software effectiveness? (1mark)

A. It removes the need for testing
B. It ensures documentation accuracy
C. It helps compare actual system behaviour with user expectations
D. It reduces development costs directly

7. Which testing type MOST effectively detects defects within individual functions before integration? (1 mark)

A. System testing
B. Integration testing
C. Unit testing
D. Acceptance testing

8. What is ORM mainly used for? (1 mark)

A. Writing raw SQL queries
B. Managing file systems
C. Interacting with databases using objects in code
D. Designing website layouts

9. A mid-sized software company needs to develop a new mobile application quickly, but its internal team lacks expertise in mobile development and is already overloaded with other projects. The company wants to reduce development time and cost.

   Which software development approach should the company choose? (1 mark)

   A. In-house Development
   B. Outsourcing
   C. Prototyping
   D. Parallel Installation

10. Which SQL statement is used to retrieve all columns from the Customers table where the Country is 'Australia'? (1 mark)

    A. SELECT * FROM Customers WHERE Country = 'Australia';
    B. GET * FROM Customers WHERE Country = 'Australia';
    C. FETCH ALL FROM Customers WHERE Country = 'Australia';
    D. SELECT ALL Customers WHERE Country = 'Australia';

11. Why might reinforcement learning outperform supervised learning in autonomous driving systems? (1 mark)

    A. It requires no data
    B. It learns optimal decisions through continuous interaction with the environment.
    C. It guarantees perfect accuracy
    D. It eliminates ethical concerns

12. A project team decides to delay implementing several features due to time constraints.

    Which type of requirement would MOST appropriately be postponed? (1 mark)

    A. Authentication system
    B. Database connectivity
    C. Error handling
    D. Optional user personalisation features

13. Which Big Data "V" MOST directly drives the need for real-time processing architectures? (1 mark)

A. Variety
B. Veracity
C. Velocity
D. Volume

14. If packets arrive out of order at the destination, which protocol ensures they are rearranged correctly? (1 mark)

A. DNS
B. HTTP
C. IP
D. TCP

15. Tick the boxes that correctly describe features of each protocol. (3 marks)

Feature	FTP	HTTP
Used to transfer web pages between a browser and a server	☐	☐
Commonly used for uploading and downloading files	☐	☐
Uses port 80 by default	☐	☐
Transfers data in plain text by default	☐	☐

16. Which of the following statements correctly describe how machine learning supports automation in modern software systems? (3 marks)

Circle all that apply.

A. Machine learning systems can improve performance over time by learning from new data
B. Machine learning replaces the need for human oversight in all automated systems
C. Machine learning can help optimise decision-making in areas such as fraud detection and recommendations
D. Machine learning automation always guarantees unbiased and accurate outcomes
E. Machine learning enables automation systems to adapt to changing patterns rather than relying only on fixed rules
F. Machine learning can function effectively without access to data

Section II

60 marks
Attempt Questions 17-33
Allow about 2 hours for this section.

Answer each question in the booklets provided. Ensure all diagrams are clearly labelled.

17. Outline one difference between AI and ML (2 marks)

18. Outline one benefit of data protection (2 marks)

19. Discuss any ONE issue related to communication in Project Management (3 marks)

20. A small food delivery business operating in a semi-urban area wants to offer customers a convenient way to view restaurant offerings, submit orders, and monitor delivery progress. A large portion of its customer base experiences inconsistent internet access and uses devices with limited storage capacity. The business is working with a restricted budget and limited technical resources, making the development and long-term maintenance of multiple platform-specific applications impractical.

    Justify the use of a PWA for the above scenario by clearly outlining at least three benefits of using a PWA. (3 marks)

21. Compare and contrast SAST and DAST. (3 marks)

22. Discuss ONE ethical issue associated with automated systems using machine learning. (3 marks)

23. A large online retail company is experiencing rapid growth. As the number of customers increases, the company is struggling to keep up with frequent software updates, high support workloads, and complex internal business processes such as order processing, inventory management, and customer service requests.

    To address these challenges, the company decides to automate some of these process. DevOps, RPA and BPA are some of the strategies considered by the company.

    Describe how machine learning supports automation within any ONE of these automation processes. (3 marks)

24. Explain the key features of privacy by design that would assist in securing any software solution. (3 marks)

25. Compare supervised learning and unsupervised learning in software automation, providing one scenario/example of where each of them could be used.

    (3 marks)

26. Explain the purpose of session management in web applications. (3 marks)

27. A community organisation is developing a digital platform to help members register for events, receive updates, and access resources. While the initial goals of the platform are clear, the organisation expects its needs to change over time as member feedback is collected and new services are introduced. Stakeholders would like to see usable features early, even if the system is not fully complete, so they can evaluate how well the platform meets user expectations. Regular input from users is important, as many members have differing accessibility needs and preferences that may not be fully understood at the start of the project.

    (i) Which software development approach is most suitable for this scenario?

    (1 mark)

    (ii) Justify your choice, referring to the scenario in your answer. (3 mark)

28. (i) Describe ‘Sandboxing’ concept in Secure Software Architecture. (3 marks)

(ii)Provide an example where sandboxing can be used to avoid security issues. (2 marks)

29. Describe the role of testing in ensuring software quality. (3 marks)

30. Complete the data dictionary below for a food delivery system. Variable names should be relevant to the food delivery system. (3 marks)

Variable Name	**Data Type**	**Size / Format**	**Description**	**Example Value**
user_id	___	50 chars	_____	1045
_____	str	50 chars	Customer’s full name	Alex Morgan
email	str	100 chars		alex@email.com
restaurant_id	int	50 chars	Unique identifier for each restaurant	22
product_name	____	80 chars	Name of the food item	Chicken Burger
price	____	decimal	Cost of the menu item	14.50
order_id	int	50 chars	Unique identifier for each order	9008
order_status	___	20 chars	Current delivery status of the order	Delivered

31. Discuss “overfitting” in Machine Learning, citing one example. (3 marks)

32. A local food delivery business operates in a busy suburban area and partners with several independent restaurants. Customers place orders through the business’s website, selecting multiple food items from a menu. Each item has a fixed price, and the system calculates the total cost of the order before payment.

    To encourage larger orders, the business offers a 10% discount when the total order value exceeds $30.

    The system must calculate the total, apply the discount where appropriate, and display a clear summary for both customers and staff.

(i) A customer places an order containing several food items. The prices of the items are stored in the list below:

prices = [12.50, 8.00, 5.75]

Write a Python program that calculates and prints the total cost of the order. (2 marks)

(ii) The business wants to automate order processing. (6 marks)

Write a Python function called process_order() that calculates the total cost of the order, applying a 10% discount if the total is greater than $30.

The function should take the following parameters: customer_name, and prices. It should return the final amount payable.

Other requirements:

- Use meaningful variable names
- Apply the discount only when the condition is met
- Round the final cost to 2 decimal places

Output should be printed in the format below:

Customer: Alex

Final cost: $27.45

33. A large logistics company uses machine learning to automate several parts of its operations. ML systems are used to predict delivery times, detect fraudulent transactions, and optimise warehouse workflows. As the system improves through learning from data, fewer manual checks are required by employees.

Some workers are concerned that continued automation will reduce job opportunities, while others believe machine learning will create new roles that focus on managing, maintaining, and improving these intelligent systems.

Discuss how the increasing use of machine learning in automation may impact the future of work.

In your answer, refer to the scenario and discuss both potential benefits and challenges for individuals and society. You may include your own informed opinion. (6 marks)

End of Exam